Candelaria, Arvin H.

CAS-05-401P

Assignment #1

3. Temperature Conversion: Convert temperatures from Celsius to Fahrenheit using the formula fahrenheit = (celsius* 9/5) + 32. Using rbind, combine the celsius_values and fahrenheit_values.

	[, 1]	[, 2]	[, 3]
celsius_values	25	10	-5
fahrenheit_values	77	50	23

```
celsius_values<-c(25, 10, -5)

fahrenheit_values<-print(celsius_values*9/5+32)

rbind(celsius_values, fahrenheit_values)
```

4. Stock Price Change: Given two vectors representing opening and closing stock prices for 5 days, calculate the daily change (closing - opening) for each day.

	opening_prices	closing_prices	daily_changes
[1,]	100	105	5
[2,]	112	108	-4
[3,]	98	95	-3
[4,]	105	107	2
[5,]	110	115	5

```
opening_prices<-c(100, 112, 98, 105, 110)

closing_prices<-c(105, 108, 95, 107, 115)

daily_changes<-print(closing_prices-opening_prices)

cbind(opening_prices, closing_prices, daily_changes)
```

1. Area of a Triangle: Given base (5) and height (8) as numeric values, calculate the area of a triangle using the formula (1/2) base height.

"Area of triangle: 20"

```
base<-5

height<-8

x<-print((1/2)*base*height)

paste("Area of triangle:", x)
```

2. Age in Dog Years: Create a vector with human ages (e.g., c(20, 35, 12)) and convert them to dog years using the formula dog_years = 15+4* human_years. Using cbind, combine the human ages and dog_years.

	human_ages	dog_years
[1,]	20	95
[2,]	35	155
[3,]	12	63

```
human_ages<-c(20, 35, 12)

dog_years<-print(15+4*human_ages)

cbind(human_ages, dog_years)
```